REPORT

1. Introduction

Employee Attrition Analysis is a data analytics project focused on identifying the factors influencing employee turnover. The project analyses employee demographic, job, compensation, satisfaction, and work-related data to identify patterns in attrition. **Power BI** is used to visualize these patterns through interactive dashboards and generate insights that can support effective employee retention and HR decision-making.

2. Dashboard

2.1 Attrition Overview

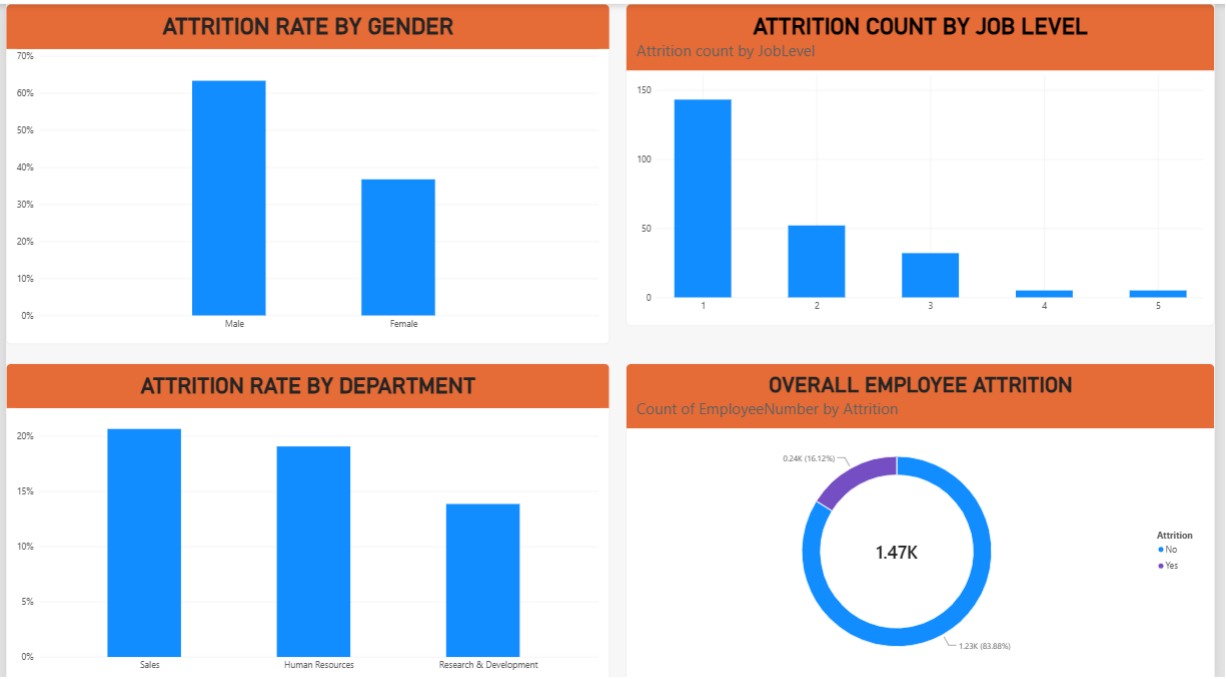


Fig 2.1 Overall Attrition Overview

This dashboard provides an overall view of employee attrition across key demographic and organizational factors. It includes overall employee attrition, attrition rate by gender, department, and job level. The dashboard helps identify which employee groups and organizational areas experience higher levels of attrition. The overall attrition chart shows that 16.12% of employees have left the organization, providing a baseline for comparison with the other factors.

2.2 Tenure and Job Role Analysis



Fig 2.2 Tenure and Job role Analysis

This dashboard focuses on the relationship between income, years at the company, job role, and years since the last promotion and employee attrition. It helps identify whether compensation levels, employee tenure, career progression, or specific job roles are associated with higher turnover. The dashboard provides a view of the employee groups that may be more vulnerable to leaving based on their career stage and position within the organization.

2.3 Educational and Demographic Analysis



Fig 2.3 Demographic and Educational Analysis

This dashboard analyzes employee attrition based on age group, job level, marital status, education field, and education level. The visualizations help understand whether demographic and educational characteristics are associated with differences in employee turnover. The dashboard highlights variations in attrition across different age groups and educational backgrounds, allowing HR teams to identify demographic segments that may require further investigation.

2.4 Satisfaction and Work Environment Analysis



Fig 2.4 Job satisfaction and Work environment

This dashboard examines workplace-related factors that can influence employee retention. It includes overtime, work-life balance, job satisfaction, and environment satisfaction. The visualizations allow comparison of attrition rates across different satisfaction levels and working conditions. This page helps identify potential workplace factors that may contribute to employee turnover and can support HR in developing employee engagement and retention strategies.

2.5 Work Experience, Travel and Training Analysis



Fig 2.5 Work Experience and Training Analysis

This dashboard focuses on employees' business travel, total working experience, distance from home, and training frequency. It examines whether work-related mobility, professional experience, commuting distance, and training participation are associated with employee attrition. The dashboard helps identify patterns in employee turnover related to work requirements, career experience, commuting factors, and training opportunities.

3. Business Insights

The analysis highlights several factors that are closely associated with employee attrition:

- The overall attrition rate is 16.12%, meaning approximately one in every six employees has left the organization.
- Overtime is a major factor, with employees working overtime showing an attrition rate of 30.53%, compared with 10.44% for those who do not.
- Younger employees (18–25 years) have the highest attrition rate at 35.77%, indicating a need for stronger early-career support and engagement.
- Sales has the highest departmental attrition rate at 20.63%, while Sales Representatives have the highest attrition among job roles at 39.76%.

- Employees earning below 3K have a considerably higher attrition rate of 28.61%, suggesting that compensation may influence retention.
- Employees who travel frequently have a higher attrition rate of 24.91%, indicating that frequent travel may contribute to employee turnover.
- Lower levels of job satisfaction and work-life balance are associated with higher attrition, highlighting the importance of a supportive work environment.
- Employees in the early stages of their careers and organizational tenure show higher attrition compared with more experienced employees.

4. Business Recommendations

- Manage overtime and workload by improving workforce planning and distributing responsibilities more effectively.
- Strengthen support for younger employees through mentoring, career guidance, training, and structured development programs.
- Focus on high-attrition roles and departments, particularly Sales and Sales Representative positions, by reviewing workload, incentives, and career growth opportunities.
- Review compensation structures, especially for lower-income employees, to ensure that pay remains competitive and aligned with responsibilities.
- Improve work-life balance through flexible work arrangements, reasonable workloads, and employee well-being initiatives.
- Increase employee engagement and satisfaction through regular feedback, recognition programs, and stronger manager-employee communication.
- Provide clearer career progression opportunities through skill development, internal promotions, and career planning.
- Review business travel requirements and provide appropriate support for employees who travel frequently.

5. Conclusion

The Employee Attrition Analysis highlights the key factors influencing employee turnover and provides actionable insights for improving retention. The findings demonstrate how data analytics and Power BI can support better HR decision-making and help build a more engaged and stable workforce.